

Skin Lesion Segmentation with GWO & U-Net

Optimizing hyperparameters for enhanced medical imaging accuracy using Gray Wolf Optimization on ISIC 2018 dataset

 REGAN AGAM
 24/552177/PTK/16439
 Universitas Gadjah Mada

 Architecture

U-Net

 Optimization

Gray Wolf Algorithm

 Dataset

ISIC 2018

 Performance Metric

Dice Score

The Challenge: Precision in Diagnosis



Early Detection

Lesion boundaries are **early indicators** of melanoma progression



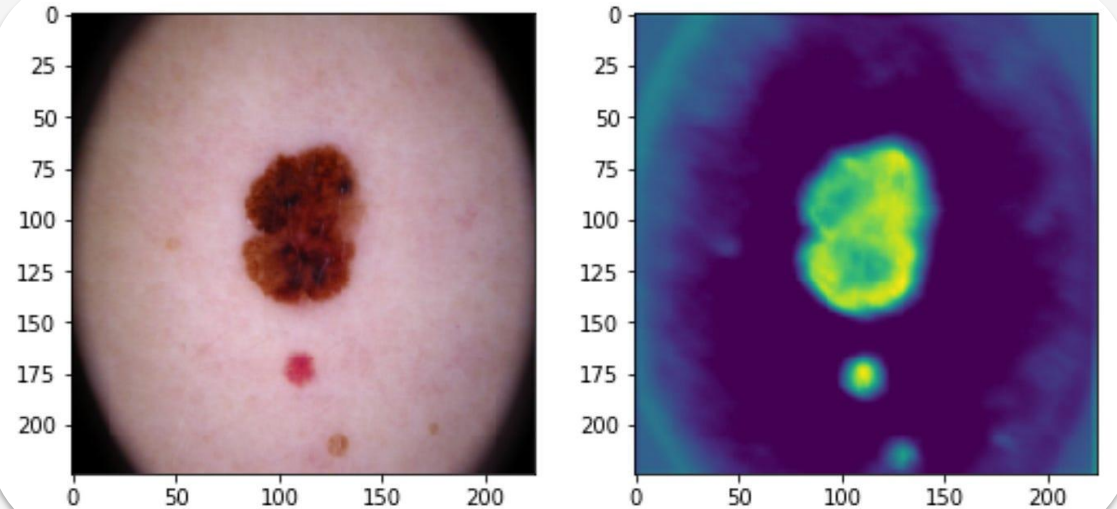
Automation Need

Manual segmentation is **time-consuming** and prone to inter-observer variability

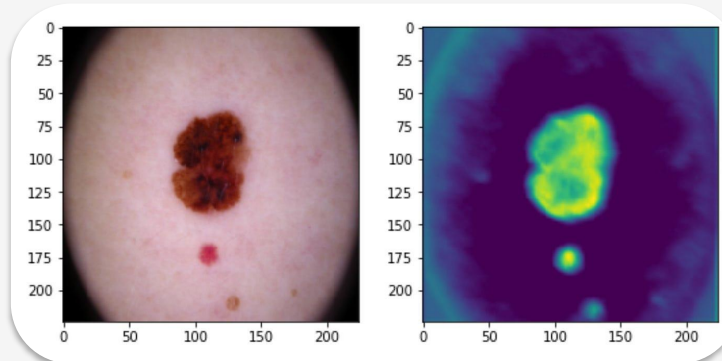
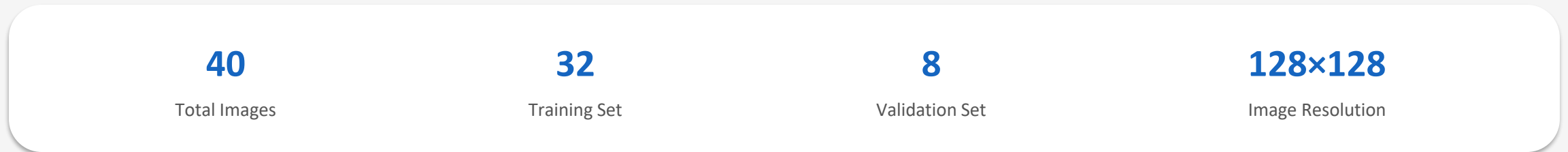
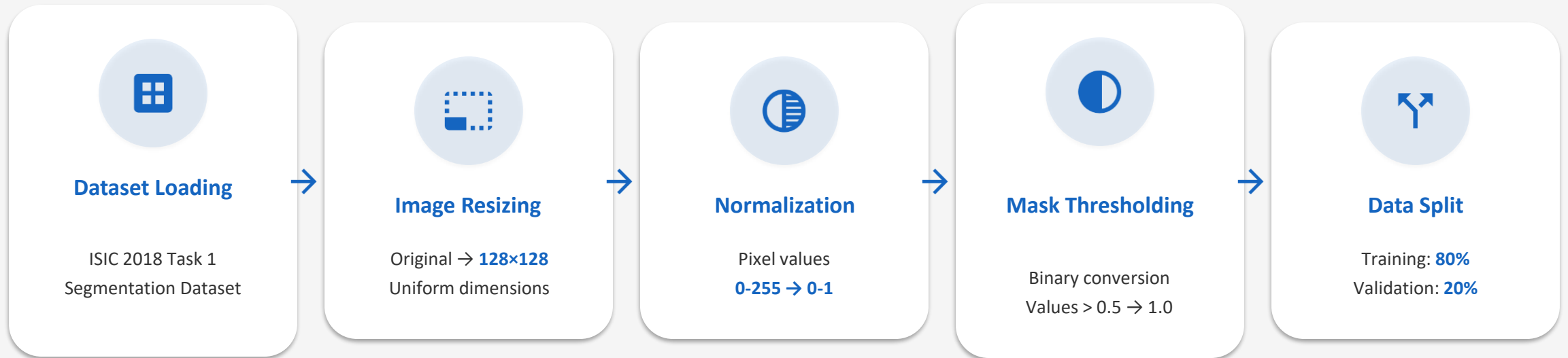


Technical Hurdles

Variable lesion appearance, irregular boundaries, and low contrast



Data Preprocessing Pipeline



Model Architecture: U-Net



Encoder-Decoder

Contracting path extracts **context features**

Expansive path enables **precise localization**



DoubleConv Blocks

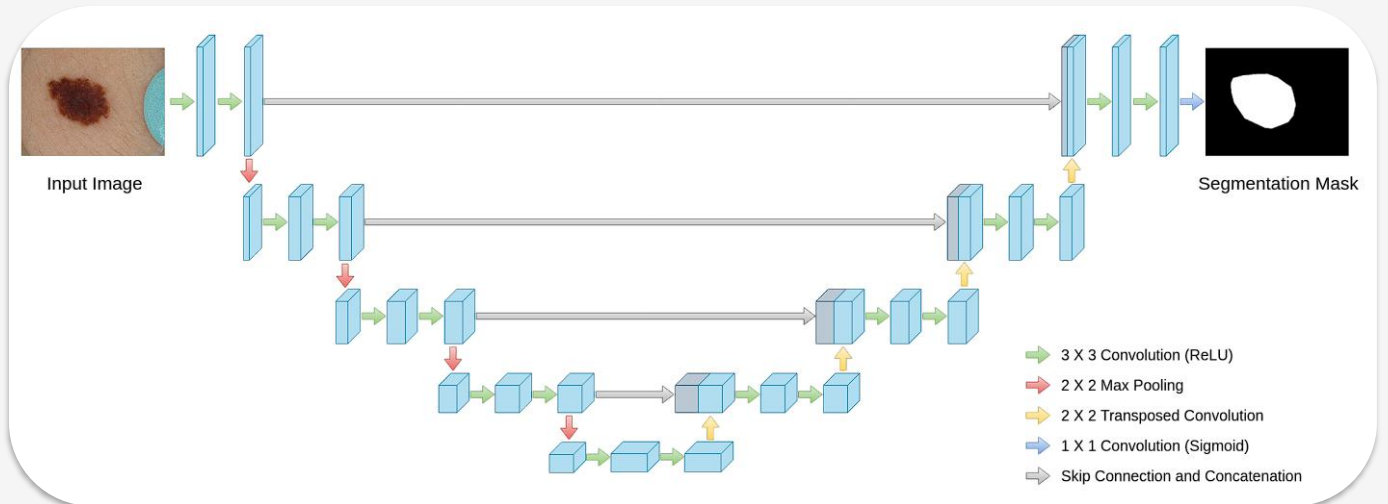
Two consecutive convolutions with **BatchNorm** and **ReLU**



Skip Connections

Concatenates features from encoder to decoder

Preserves **spatial details** for accurate segmentation



Gray Wolf Optimization



Metaheuristic Algorithm

Inspired by **wolf pack hunting behavior**

Optimizes hyperparameters through **social hierarchy** and **cooperative hunting**



Hunting Mechanism

Wolves surround prey with **decreasing radius**

Parameter **a** controls exploration vs exploitation

α (Alpha)

Best solution

β (Beta)

Second best

δ (Delta)

Third best

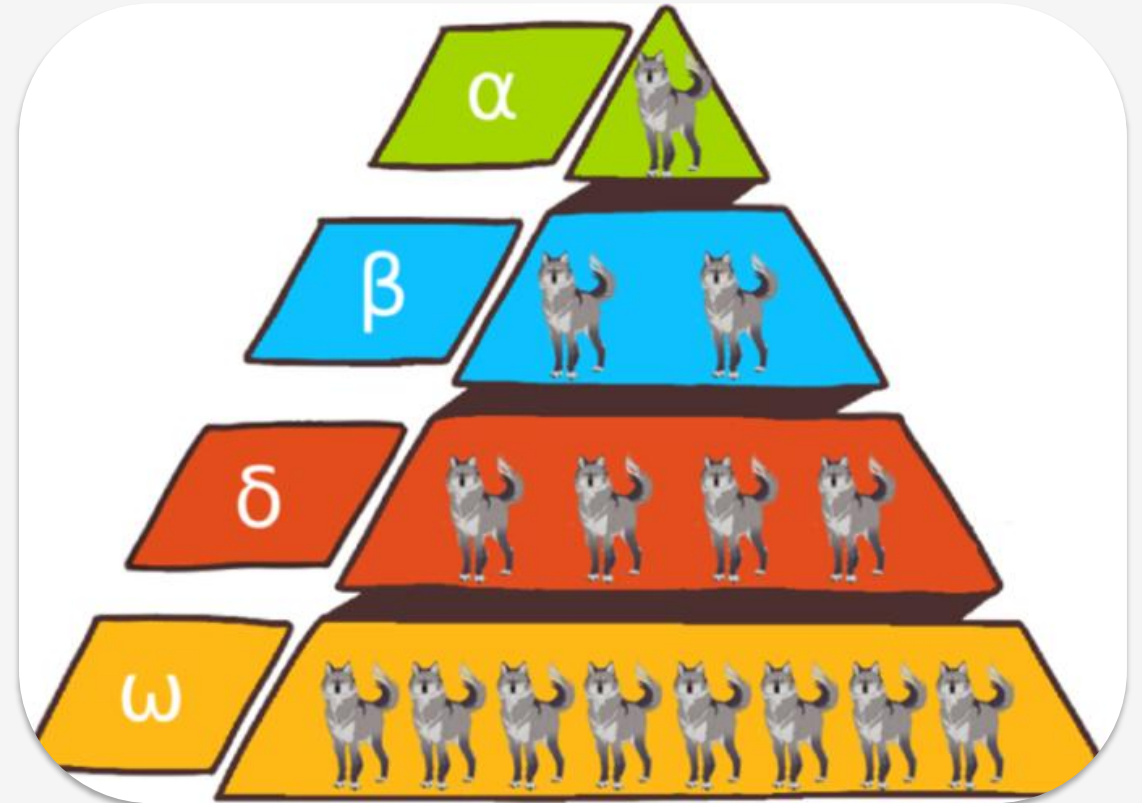
Optimized Hyperparameters

Learning Rate

0.0001 - 0.01

Weight Decay

1e-5 - 0.001



Experimental Setup



Baseline U-Net



Learning Rate: 0.001



Weight Decay: 1e-5



Optimizer: Adam



Loss Function: Dice Loss



GWO-Optimized U-Net



Population Size: 3 wolves



Iterations: 6



Optimal LR: 0.007347



Optimal Decay: 0.000603

Common Training Settings

15

Training Epochs

4

Batch Size

Dice

Evaluation Metric

ISIC 2018

Dataset

Baseline U-Net Results

Training Performance

Epoch 5 Loss

0.4555

Epoch 10 Loss

0.3469

Epoch 15 Loss

0.3171

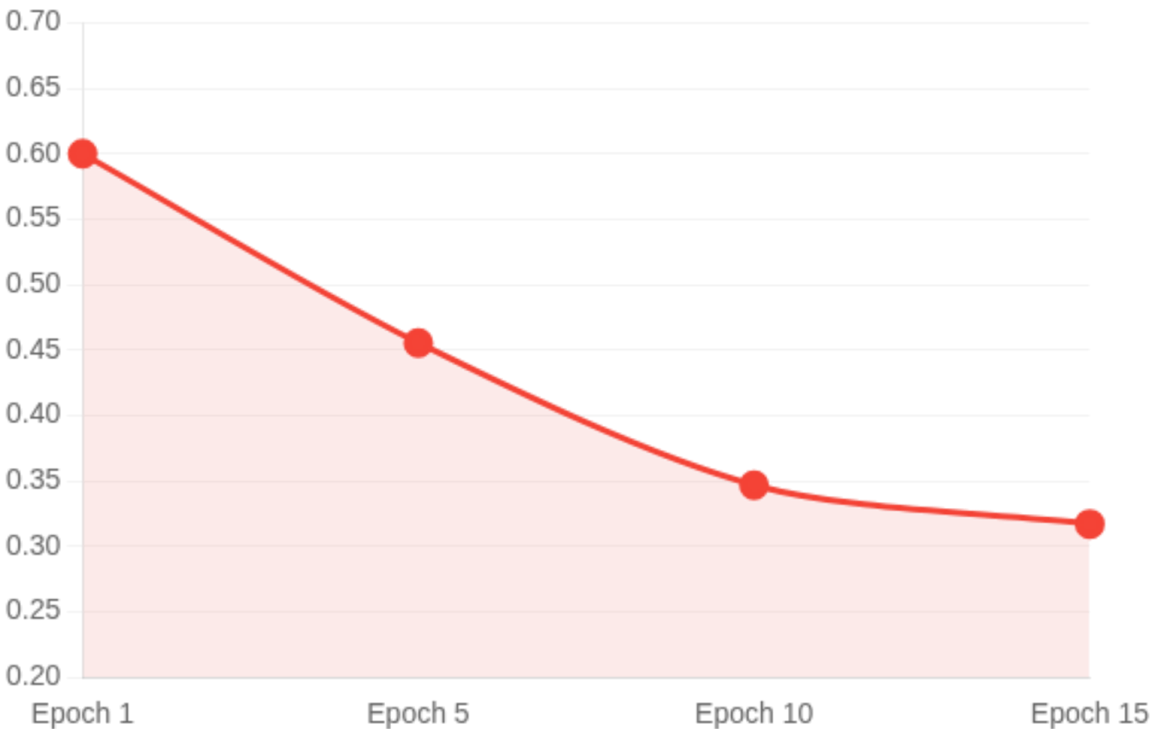
Loss Reduction

30.4%

Dice Score on Test Image

0.76

Training Loss Progression



Learning Rate

0.001

Weight Decay

1e-5

Optimizer

Adam

Loss Function

Dice Loss

GWO-Optimized U-Net Results

Winner

Training Performance

Epoch 5 Loss

0.2221

Epoch 10 Loss

0.2177

Epoch 15 Loss

0.1760

Loss Reduction

20.7%

Dice Score on Test Image

0.95

 **25 % better than baseline**

Loss Progression Comparison



Optimal LR

0.007347

Optimal Decay

0.000603

Optimizer

Adam

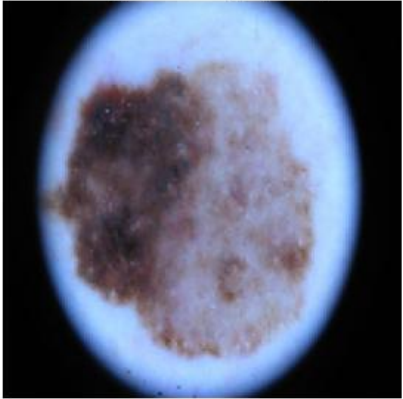
Loss Function

Dice Loss

Single Test Image Comparison

Original Image

Original Image (ISIC)



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 ISIC 2018 Sample

Ground Truth

Ground Truth Mask



🔗

 Reference Mask

Baseline U-Net

Baseline Prediction (Dice: 0.76)



Dice: 0.76

📉 Lower Performance

GWO-Optimized

GWO Prediction (Dice: 0.95)



🏆 Winner

Dice: 0.95

📈 Better Performance

Single Test Image Performance

0.76

Baseline U-Net Dice Score

0.95

GWO-Optimized U-Net Dice Score

📈 17.3% Improvement

Conclusion

GWO Wins



Key Findings

GWO-Optimized U-Net **outperformed** baseline model

Baseline U-Net

0.76

GWO-Optimized

0.95



25% improvement on test image



GWO Advantages

Optimal hyperparameters found automatically

Learning rate: **0.007347** vs baseline 0.001

Weight decay: **0.000603** vs baseline 1e-5

Better exploration of parameter space

Lower final loss: **0.1760** vs 0.3171

Thank You



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